Supplementary Material

A Geometric Criterion for Sway and Rocking in Inclined Pile Groups, with Field Identification of an Offshore Wind Turbine Foundation

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This document collects the material that supports the main paper but is not needed to follow its argument: the quality of the raw records, the simulation that sets the threshold of the first check, the complete list of identified poles, the test of the processing settings, the search for a tower mode, the setup-by-setup comparison of the two campaigns, and the sensitivity of the centre of rotation to the tower. Section, figure and table numbers without the prefix S refer to the main paper.

S1 Quality of the raw records

Table S1 gives the statistics of the raw records of the seven setups, and Figure S1 shows a typical record of campaign 2. At one moment a single channel jumps far above its normal level while the others stay quiet. The event rings briefly at a high frequency, as a sensor mounting does when it is struck, and is most likely a knock on or near that sensor rather than a motion of the cap, which would appear on every channel. Most of the signal energy lies in the mains harmonics between 50 and 200 Hz, and less than 1 % in the structural band below 5 Hz. The records of campaign 1 are less steady still: the signal level of some channels changes strongly during a record, and some channels are much weaker than others. Only the three-channel setup of campaign 1 is close to steady broadband noise.

Table S1: Measurement setups and raw-record statistics. *Ch.* is the number of channels and *Min.* the record length in minutes. *Peak*/σ is the largest absolute sample of any channel divided by that channel's standard deviation; *Kurt.* the largest channel kurtosis; $N_{>8\sigma}$ the largest number of samples of one channel beyond 8σ; *Drift* the ratio of largest to smallest channel RMS over ten equal segments; *Spread* the ratio of the largest to the smallest channel standard deviation within a setup. The largest value of each column is in bold

Camp.	Setup	Ch.	Samples	Min.	Peak/σ	Kurt.	$N_{>8\sigma}$	Drift	Spread
1	Vertical	5	2 535 936	25.6	133	562	2238	3.2	6.1
1	Lateral 1	5	1 523 873	15.4	25	54	**3131**	12.9	**47.0**
1	Lateral 2	5	1 385 189	14.0	50	1812	1063	**112**	9.2
1	Three-channel	3	1 545 336	15.6	10	4	9	1.5	33.7
2	Lateral	6	906 399	9.1	**139**	**3883**	724	4.7	11.5
2	Vertical d1	6	1 875 536	18.9	69	95	695	4.4	4.1
2	Vertical d2ke	6	2 260 219	22.8	67	88	2564	4.3	4.2

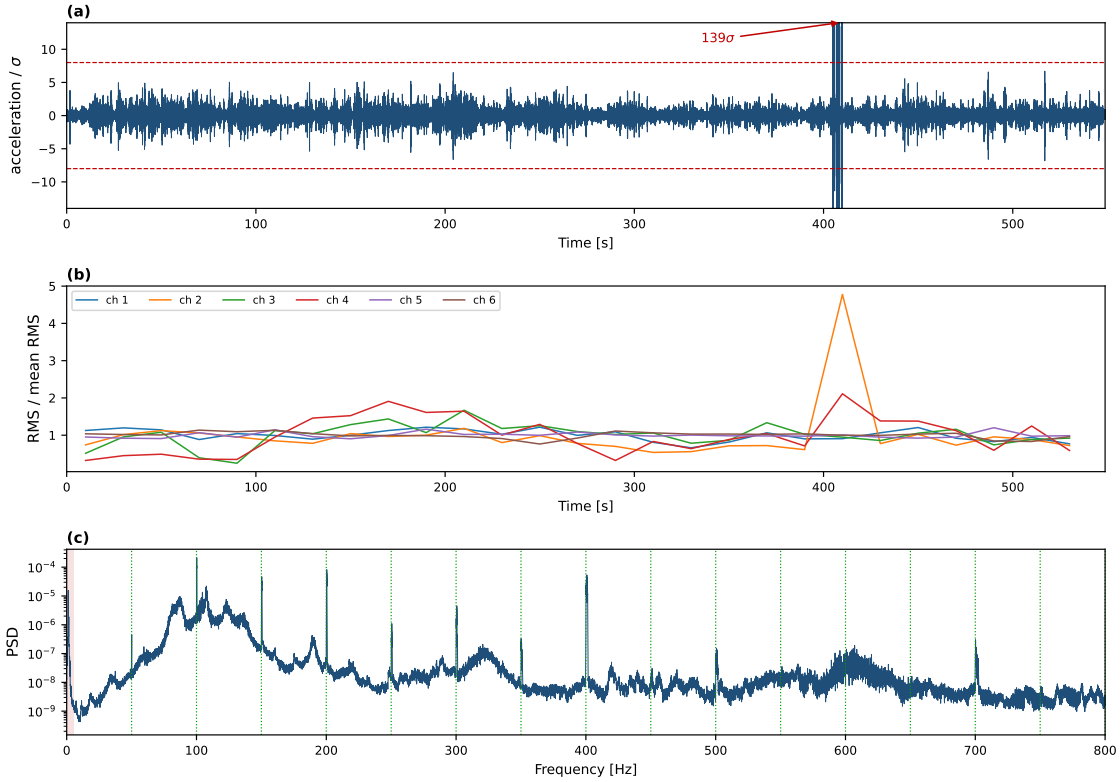


Figure S1: One record of campaign 2. (a) Time history of the channel with the largest event (arrow), in units of its standard deviation. (b) Running RMS of every channel; the event appears on one channel only. (c) Spectrum averaged over the channels, with multiples of 50 Hz dotted and the structural band shaded

S2 Threshold of the effective channel count

The first check of Section 4.2 rejects a pole whose effective channel count $N_{\mathrm{eff}} = (\sum |\phi|^2)^2 / \sum |\phi|^4$ is below 1.25. The threshold was set by simulation. Two cases were generated with the sampling rate, the record length of the shortest record (28 000 samples at 51.61 Hz) and the spectral resolution of the measurements:

- *local response*: a narrow-band signal at 1.2 Hz with 2 % damping on one channel only, and independent white noise on every channel;
- *mode of the cap*: the same signal shared by all five or six channels with the measured shape of the low band, in which opposite corners move in antiphase and the centre hardly moves, again with independent noise on every channel.

Each case was repeated 400 times, at signal-to-noise ratios of 3 and 10, and N_{eff} was computed exactly as for the measurements, from the first singular vector of the cross-power spectral density matrix at the peak of the 1.0–1.4 Hz band. A signal on a single channel never gave more than 1.00. A mode shared by the channels gave a median of 1.55 with five channels and 1.77 with six, and 1.5 or more in 95 % of the trials, at either signal-to-noise ratio. The threshold of 1.25 was placed between the two cases; it is a convention, not an optimised bound. The shape used for the shared mode is the measured shape of the vertical setup of campaign 1, so the simulation tests the effect of noise and level, not of shape. In the measurements N_{eff} is evaluated once for each setup and band, at the spectral peak of the band, not for each pole.

Two sensitivities were also examined. A difference of gain of up to 3 % between channels, the largest difference between the sensitivities listed in the processing log, changes N_{eff} of the shared

mode by at most 4 % (1.49 to 1.62), which leaves it well above the threshold. Channel levels that differ by up to a factor of ten are more harmful: the count of a real mode then fell below 1.25 in 21 % of the trials, so the check can wrongly reject a real mode in such a record. In the same trials a local response never exceeded 1.01, so the check did not accept one. With three channels the two cases cannot be told apart, and the threshold is not applied to the three-channel setup.

S3 Complete list of identified poles

Table S2 lists every pole returned below 25 Hz, by setup, with the spectral peak of frequency domain decomposition (FDD) nearest to the low-band poles. Poles with damping outside 0.1 to 15 % are discarded by the identification and do not appear. Table S3 gives the phase between opposite corners of the instrumented frame at the peak of the low band, which underlies Figure 7 of the main paper.

Table S2: Poles returned by the identification, by setup; frequency in Hz with damping ratio in per cent in parentheses. *FDD peak* is the spectral peak of frequency domain decomposition in the low band (Hz); lateral setup 2 has none there

Camp.	Setup	Poles below 3 Hz	FDD peak	Poles from 3 to 25 Hz
1	Vertical	1.174 (4.74)	1.134	18.132 (0.70), 18.565 (0.68), 19.077 (0.58), 19.466 (0.59), 19.920 (1.30)
1	Lateral 1	1.226 (3.43), 1.254 (2.77), 1.731 (5.49)	1.248	21.920 (0.40), 22.751 (0.51), 23.754 (0.30)
1	Lateral 2	1.322 (2.71)	none	21.574 (5.95), 22.813 (1.25), 23.173 (0.59)
1	Three-channel	1.238 (3.01), 1.331 (4.39), 1.726 (5.35)	1.235, 1.311	14.438 (1.42), 14.996 (0.97)
2	Lateral	1.052 (2.69), 1.215 (1.93), 1.612 (2.42)	1.197	none
2	Vertical d1	1.147 (9.85)	1.210	11.806 (11.52), 13.871 (3.02), 16.833 (2.73), 17.087 (0.59), 17.673 (1.96), 18.954 (1.08), 19.473 (1.37), 19.689 (1.11), 20.549 (0.54)
2	Vertical d2ke	1.074 (1.25), 1.199 (2.35), 1.231 (2.48), 2.296 (1.19)	1.235	none

Table S3: Phase between opposite corners of the instrumented frame at the low-band peak, and amplitude at the centre relative to the mean of the corners. In the lateral setup, channel 4 is the only vertically oriented channel, which is why the 2–4 pair behaves unlike the 1–3 pair there. The peak is taken from a spectrum with twice the frequency resolution of the FDD of Table S2, so the two peak frequencies differ by up to 0.006 Hz

Setup	Direction	Peak (Hz)	1 vs 3	2 vs 4	Centre / corners
Vertical, campaign 1	vertical	1.134	−179.1°	−173.3°	0.048 / 0.466
Vertical d1, campaign 2	vertical	1.210	+176.3°	+176.6°	0.424 / 0.517
Vertical d2ke, campaign 2	vertical	1.229	−142.7°	−175.2°	0.183 / 0.518
Lateral, campaign 2	horizontal	1.203	−4.9°	−143.3°	0.927 / 0.757

Poles outside the two main groups. A few poles belong neither to the 1.05–1.33 Hz band nor to the 14–24 Hz set, and are not interpreted. A pole at 1.61–1.73 Hz appears in the three setups with horizontal channels other than lateral setup 2, which failed the first check (lateral 1,

three-channel, and lateral of campaign 2), with damping of 2.4 to 5.5 %; it cannot appear in the all-vertical setups if it is a horizontal or torsional motion, and it is left unclassified. In the lateral setup of campaign 2 its central channel carries 15 % of the corner amplitude. Single poles at 2.30, 11.8 and 13.9 Hz appear in one vertical setup each.

Agreement between the vertical setups of campaign 2. The two vertical setups of campaign 2 measure the same five points. The pole of setup d1 at 1.147 Hz has a modal assurance criterion (MAC) of 0.97 with the pole of setup d2ke at 1.074 Hz, 6 % lower in frequency, and of 0.34 and 0.44 with its poles at 1.199 and 1.231 Hz, whose shapes are strongly complex (modal phase collinearity 0.44 and 0.50). The shapes at the two spectral peaks have a MAC of 0.09. The band therefore contains more than one mode, each setup is dominated by a different member, and five or six channels do not resolve the individual members. A high MAC is not by itself evidence of a mode of the cap: the 14–24 Hz poles of the two lateral setups of campaign 1 reach 0.996 to 0.998 because one channel dominates both.

S4 Processing settings and the mains harmonics

Subspace identification builds its model from the covariances of the records over a time window $\tau = i\,\Delta t$, set by the number of block rows i. At the acquisition rate of 1651.6 Hz with $i = 150$, the window is only 0.09 s, a tenth of a period at 1 Hz. The effect of this window and of the high-pass cut-off was tested on one record (Table S4 and Figure S2). With the short covariance window of the full rate, no pole is found below 5 Hz, whether or not the high-pass filter is applied (configurations A and C). The 1 Hz band appears only with both the longer window and the low cut-off (configuration B), which is the configuration used in the main paper. Configuration E, which keeps the lower sampling rate but shortens the window to 0.097 s, returns no pole at all, so the window, not the sampling rate, is what matters.

Table S4: Comparison of processing configurations, one record (lateral setup, campaign 2). Configurations A–D form a factorial in window and high-pass cut-off, in which the window changes together with the sampling rate; E differs from B in the window alone

Config.	Sampling rate	High-pass	i	τ	Poles returned
A	1651.6 Hz	5 Hz	150	0.091 s	20, from 57.6 to 700.9 Hz; none below 5 Hz
C	1651.6 Hz	0.3 Hz	150	0.091 s	19, from 80.9 to 604.6 Hz; none below 5 Hz
D	51.6 Hz	5 Hz	120	2.33 s	6, from 6.8 to 20.6 Hz; none below 5 Hz
B	51.6 Hz	0.3 Hz	120	2.33 s	3: 1.052, 1.215, 1.612 Hz
E	51.6 Hz	0.3 Hz	5	0.097 s	none

Seven of the poles returned at the full rate have damping between 0.007 and 0.042 %, far below the 0.1 % bound of the third check. All seven lie within 0.2 % of a multiple of 50 Hz, at 100.17, 150.30, 200.28, 250.26, 300.49, 400.69 and 700.93 Hz. They are harmonics of the mains supply. For this test the lower damping limit of the identification was relaxed from 0.1 to 0.001 %; with the limit of the main analysis these poles would have been discarded. The other thirteen poles of configuration A were not classified, since that configuration is not used.

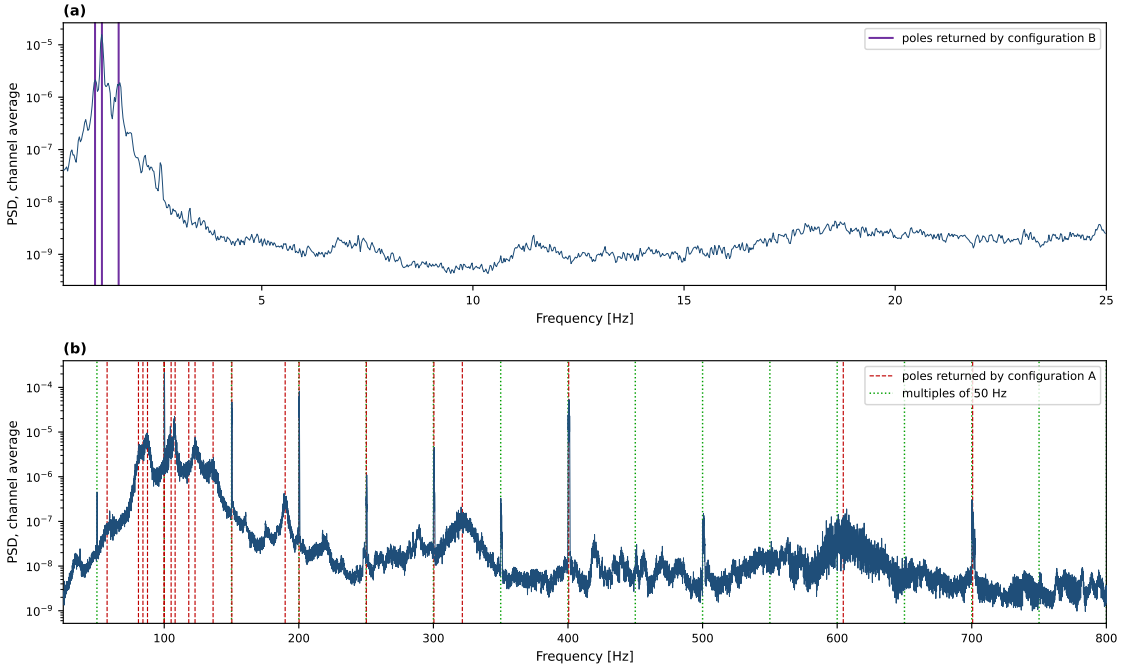


Figure S2: The same record seen by configurations A and B of Table S4, over (a) the structural band from 0.3 to 25 Hz and (b) the band from 25 to 800 Hz from which configuration A draws its poles. Dotted lines mark multiples of 50 Hz

S5 Search for a tower mode below 0.5 Hz

The band below 0.5 Hz was searched for a tower mode, which a soft–stiff design places near 0.3 Hz [1]. The records were decimated further to 12.9 Hz and analysed with a covariance window of 11.6 s (Figure S3). Four of the seven records were searched: the vertical setup and lateral setup 1 of campaign 1, and the lateral setup and vertical setup d2ke of campaign 2. No pole below 0.5 Hz passes the checks. The only features are spectral peaks near 0.21 Hz in campaign 1, recorded before the tower stood and consistent with waves, and near 0.32 Hz in campaign 2, where the first tower mode is expected.

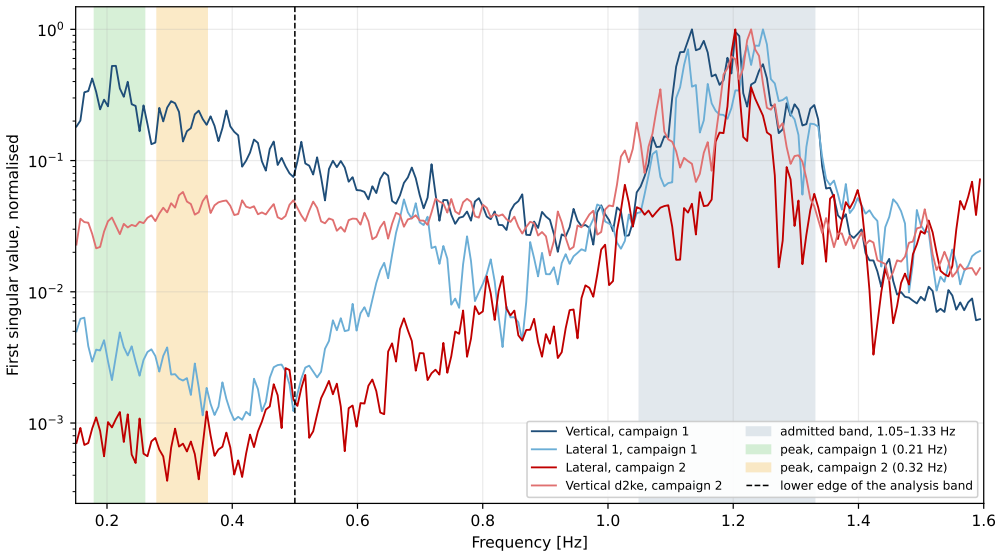


Figure S3: Search below the analysis band, with the records decimated to 12.9 Hz and a covariance window of 11.6 s. The only features below 0.5 Hz are the peaks marked at 0.21 Hz in campaign 1 and 0.32 Hz in campaign 2. The first, recorded before the tower stood, is consistent with wave excitation; the second lies where the first tower mode is expected. Neither yields a stable pole

S6 Setup-by-setup comparison of the two campaigns

Table S5 lists the low-band poles behind Figure 9 of the main paper. Averaging all setups of each campaign, including lateral setup 2, suggests a fall of 8.4 % (6.8 % without it) after the tower was installed, but this mixes setups that measure in different directions and therefore weight different members of the band. The table compares setups that measure in the same direction.

The differences between setup means hide a larger scatter. Pole by pole, the three poles of vertical setup d2ke differ from the single pole of the vertical setup of campaign 1 by −8.5, +2.1 and +4.9 %, and the 1.215 Hz pole of campaign 2 differs from the two poles of lateral setup 1 by −0.9 and −3.1 %. Between the three segments of one record a pole moves by up to about 5 % (vertical setup d1: 1.163 and 1.104 Hz, with no pole in the third segment). The spectral peaks (Table S2) differ from the subspace poles by up to about 5 % and, in the vertical setups, rise from campaign 1 to campaign 2 (1.134 to 1.210 and 1.235 Hz) where the subspace poles fall. The sign of any change is therefore not determined.

Orientation of channel 4 in campaign 1. The channel orientations of the lateral setups of campaign 1 are not recorded. In lateral setup 1 the amplitude of channel 4 is 0.71 of the mean of the other channels, with a coherence of 0.94. This is what a fifth horizontal channel would give, and the setup is treated as horizontal throughout; it is not used for the ratio of vertical to horizontal motion.

Table S5: Low-band poles by setup, grouped by measured direction, and the change between campaigns for like-for-like setups. V: all channels vertical; H: horizontal, with channel 4 vertical in campaign 2; M: one vertical and two horizontal channels

Dir.	Camp.	Setup	Poles (Hz)	Setup mean (Hz)
V	1	Vertical	1.174	1.174
V	2	Vertical d1	1.147	1.147
V	2	Vertical d2ke	1.074, 1.199, 1.231	1.168
		change, vertical: $-2.3\,\%$ and $-0.5\,\%$		
H	1	Lateral 1	1.226, 1.254	1.240
H	1	Lateral 2 (failed check 1)	1.322	1.322
H	2	Lateral	1.052, 1.215	1.134
		change, 1.215 Hz pole against lateral 1: $-2.0\,\%$		
M	1	Three-channel	1.238, 1.331	1.284

S7 Sensitivity of the centre of rotation to the tower

The bound on the ratio of vertical to horizontal motion (Eq. 2 of the main paper) rests on a centre of rotation at or above the focal points, which the rigid-cap model gives for a cap without superstructure. The ratio was measured in campaign 2, with the tower standing. The design report gives only a lumped mass of 256 t at the top of an 88 m tower and a vertical load of 4681 kN at the tower base, about 477 t for the complete turbine; the stiffness of the tower is not given, and whether the nacelle and rotor were in place during campaign 2 is not recorded. The tower is therefore not modelled as such. Instead, one bending mode of the tower is represented by a single oscillator: a modal mass m_t at height z_t above the pile heads, connected by a spring to the point of the rigid cap body at that height, with a fixed-base frequency f_t. The oscillator adds one degree of freedom to the six of the cap. The cap model takes the central values of its parameters: an axial stiffness of 3.5×10^8 N/m per pile and a depth to fixity of 6 pile diameters, for which the cap alone has its fundamental mode at 1.25 Hz with the centre of rotation 54.7 m above the pile heads. The mass was varied from 100 to 800 t, the height from 30 to 90 m and the frequency from 0.2 Hz to a rigid connection, 120 combinations in all, which include the masses and the height of the design report. Table S6 gives the centre of rotation and the ratio at a corner of the instrumented frame for the mode in which the cap carries most of the kinetic energy.

Table S6: Centre of rotation of the cap-dominated lateral mode and ratio of vertical to horizontal motion at a corner of the frame (3.5 m from the rotation axis), with one tower mode represented by an oscillator of modal mass m_t at height z_t and fixed-base frequency f_t. The geometric bound of the main paper is 0.09

Case	Centre of rotation	Vertical / horizontal
No tower	54.7 m	0.070
Soft tower mode, $f_t \leq 0.5$ Hz; every mass and height	53.6–59.0 m	0.064–0.071
Rigid tower, mass at 30 m; 100–800 t	55.4–61.7 m	0.061–0.069
Rigid tower, mass at 60 m; 100–800 t	49.8–54.3 m	0.070–0.077
Rigid tower, mass at 90 m; 100–400 t	38.4–51.2 m	0.075–0.103
Largest ratio in the 120 combinations (400 t at 90 m, $f_t = 3$ Hz)	35.4 m	0.11

Three regimes appear. A tower mode well below the fundamental frequency of the cap, such as a first bending mode near the 0.32 Hz peak of Section S5, is isolated from the cap: its mass does not follow the motion, and the centre of rotation and the ratio are unchanged for every mass and height examined. A tower that follows the cap rigidly acts as a mass at its height: below the centre of rotation it raises the centre, near the centre it has little effect, and above the centre it lowers it. A mass of 200 t at 90 m brings the centre down to 47 m, just above the focal points, and 400 t at 90 m brings it to 38 m, below them, with a ratio of 0.10. Between these regimes, when a tower mode lies near the fundamental frequency of the cap, the two modes couple, the cap carries as little as 54 % of the kinetic energy, and no single mode can be called the mode of the foundation; with 800 t at 90 m this is the case even for a rigid tower.

The ratio does not exceed 0.11 in any of the 120 combinations, against 0.7 or more for rocking. The separation of sway from rocking therefore does not depend on the tower. The bound of 0.09 itself can be exceeded slightly, in 5 of the 120 combinations, all with a stiff tower mode and a modal mass of 400 t or more. The study uses one oscillator and one set of foundation parameters; it shows the size and the direction of the effect, not the behaviour of the actual tower.

References

[1] Arany L, Bhattacharya S, Macdonald J, Hogan SJ (2017) Design of monopiles for offshore wind turbines in 10 steps. Soil Dynamics and Earthquake Engineering 92:126–152. `https://doi.org/10.1016/j.soildyn.2016.09.024`